Master in Economics/Economia

THE EFFECT OF CRIME SHOCKS ON INTERNATIONAL TRADE FLOWS

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ABSTRACT

This Master Thesis analyzes the effect of violent crime on international trade flows using a panel data of countries from the World Bank's World Development Indicators. Homicide rates are used as a proxy for criminal violence, while countries are classified into different crime environments according to the intensity of violence. The empirical strategy relies on panel-data techniques, including static and dynamic fixed-effects models with year fixed effects. The Hausman test supports the use of fixed effects, and robust clustered standard errors are employed to address heteroskedasticity and serial correlation

The results indicate that higher homicide rates are associated with lower trade flows. Furthermore, the impact of crime varies across crime environments, suggesting heterogeneous effects of violence on international economic activity. The dynamic specification reveals substantial persistence in trade performance over time, while marginal-effects analysis shows that increases in violence generate stronger trade penalties in relatively peaceful countries.

Overall, the findings suggest that crime represents an important obstacle to international trade and economic integration. Beyond its social consequences, criminal violence imposes economic costs by reducing trade performance and increasing uncertainty for economic agents. The study contributes to the literature on crime and economic development by providing cross-country evidence on the relationship between security conditions and international trade.

Key words: Crime, International trade, Homicide rate, Fixed effects, Panel data.

JEL:C23,F14,K42,O17

1. INTRODUCTION

International trade has become one of the main engines of economic growth in the global economy; the expansion of trade flows has allowed countries to specialize, increase productivity, and deepen integration. However, this activity is challenging due to issues associated with criminal activity.

Criminal activities tend to introduce risk and uncertainty in firms, which increase internal costs. These costs include money spent by the state to maintain effective law order, as well as money spent on health personnel to deal with injuries thar might arise from violent crimes Brown & Hibbert (2017). In this context, understanding the relationship between crime and international trade is essential for evaluating the broader economic consequences of insecurity

While existing literature has extensively analyzed the role of institutional quality, governance, and political stability in shaping trade flows, relatively less attention has been given to the direct impact of crime rates. Moreover, most studies assume the homogeneous effect of crime across countries. In this work, we analyze the effect of homicide rates on trade. Overlooking the possibility that the impact of crime may differ depending on the level of violence within each country.

This study aims to address how homicide rates affect international trade flows and whether this effect varies across different crime environments. Specifically, countries are classified into four groups according to their level of crime activity based on the United Nations Office on Drugs and Crime (UNODC), allowing for the estimation of heterogeneous effects of crime on trade.

To achieve this objective, this project employs a panel data approach using country-level data over time. Fixed effects models are estimated to control for unobserved heterogeneity, and both static and dynamic specifications are considered. Additionally, interaction terms between homicide rates and crime-level classifications are included, and marginal effects are computed to facilitate interpretation.

The main contribution of this study is twofold. First, it provides empirical evidence on the relationship between violent crime and international trade using a comprehensive panel dataset. Second, it highlights the importance of considering heterogeneity in crime levels, showing that the impact of crime on trade is not uniform across countries.

Finally, the papers are structured as follows. Section 2 reviews the relevant literature. Sections 3 describe the data and methodology. Section 4 presents empirical results. Section 5 conclude and political recommendations.

2. LITERATURE REVIEW

Recent studies have explored the relationship between crime and international trade flows. Crime may affect trade through multiple channels. Empirical evidence generally suggests that higher levels of violence negatively affect exports and trade integration.

The works from Mirza & Verdier (2008) analyzed two-way interactions between trade and transnational terrorism through a structural gravity model using Log-Linear regression (OLS). Their findings reveal that criminal activity acts as a tax on trade by increasing transaction costs and security spending. While homicides reduce trade, recognizing the endogenous relationship that may exist between terrorism and bilateral trade flows.

Morales & Ricca (2019) provide a crucial case study on how the escalation of violence in Mexico directly impacted international trade flows. Using gravity model through PPML for gravity estimations and OLS with fixed effect found that when homicides increase around 10% led to a roughly 0.5% to 1% decrease in trade flows. Importantly, they note that smaller firms are more severely affected because cannot afford the expensive private security, and this vulnerability extends to logistic and export capacity.

Anderson et al. (2022)further examine this relationship through gravity model applied to 202 countries from 2000 to 2016.Using a two-stage structural approach with PPML (Poisson Pseudo-Maximum Likelihood) in the first stage to extract exporter and importer-time fixed effects, and OLS in the second stage to regress those fixed effects on homicides rates. They find that homicide rates have a negative and statistically significant impact on trade. Their estimates suggest a 5,3% decrease in market access for exporters and 8,3 decrease for importers, with particularly strong effects on differentiated manufactured goods and services.

Moreover, Bisca (2024)explain the “crowding out” effect that occurs when criminal rate are high in Latin American and Caribbean. Using macro-level panel data through panel fixed effects found that crime is significant in reducing development. They find that crimes force to divert capital from productive investment to security spending, acting as deterrent to FDI and indirectly affect exports.

Blanco et al. (2019) demonstrate that violence does not affect all types of trade equally. Using panel data from Latin America, covering 1996-2010. They find that homicide rates reduce FDI in the secondary (manufacturing) sector. While organized crime significantly deters trade in the tertiary (services) sector. In contrast, the primary sector is largely unresponsive to crime

The study of Bandyopadhyay & Sandler (2014) uses a two-good, two-factor, small open economy model to show that terrorism can either reduce or raise trade depending on critical factors. Such as the impact of crime on the intensive factors of the export or the import sector. The authors found thar a nation’s adjustment of its security level in response to greater criminality threat may moderate the impact on 30% on trade similar if a country implement tariff.

Based their study Ma et al. (2025) examine the reverse direction of this relationship, studying how export slowdowns drive crime in China cities. Using a shift-share instrumental variable approach, they find that cities experiencing more severe export slowdowns see higher crime rates. The effects are most pronounced in manufacturing-specialized regions with high concentration of workers and baseline estimation suggests that approximately 13% of variation in crime rates across cities can be explained by differences in exposure to export slowdown.

Pan & Xun (2025) similarly provide evidence of the crime-producing effect of economic shocks. Using data panel from Chinese cities with year fixed effects for causal identification, they report that severe export slowdowns can cause an increase in crime, with effect is most pronounced in manufacturing hubs with high concentrations of young migrant workers.

LaFree & Jiang (2023) studied complementary direction, examining the impact of international trade on national homicide rates using panel data with OLS regression. Their results suggest that trade globalization is significantly associated with lower homicide rates, indicating that international economic integration fosters more peaceful domestic environments.

On the other hand, Rozo (2018) takes a firm-level perspective in Colombia, using plant-level panel data to estimate how violent crime affects market price and firm behavior in Colombia. Higher violence triggers output price fall, which drives firms to reduce production, and some firms ultimately exit the market.

Rutar (2026) adopts a long-rural global perspective, using a panel of countries from 1980 to 2001 with correlational panel regression and occasional instrumental variables. While reporting negative associations between open markets and homicide. Their study ultimately finds no robust evidence that free-market reforms or globalization reduce homicide on average, concluding that recent claims about the violence-reducing effects of markets are fragile.

Levchak (2015) estimates the effect of societal conditions and indicators of globalization on national homicide rates using World bank data from period of 1980 to 1990 and multi-year with econometric techniques. The results reveal that net investment influences rates on homicide rates across several models, suggesting that trade and economic dominance may un some contexts aggravate homicide.

Furthermore, the empirical trade literature consistently highlights the importance of including controls for institutional quality, economic performance and structural factor such FDI to obtain unbiased estimates. The following studies provide a theoretical and empirical basis for the control variables employed in the present work.

Loncan (2024) examines the effects of violence crime on corporative investment and financing decision of Brazilian Firms, exploring city-level variation in homicides. Using a dynamic corporate investment regression model with year fixed effects to absorb economic cycles, the study finds that homicides affect investment efficiency and financing choices, decoupling investment from debt finance and profitability. The negative association between homicides and investment is significantly stronger in small firms, underscoring the uneven costs that violent crime imposes, with downstream effects that reduce trade capacity.

Other papers such a Garriga & Phillips (2023) examine organized crime and FDI in Mexican states between 2000 and 2018. This is a particularly relevant case given that Mexico is one of top global recipients of FDI. Their results indicate deter trade in Mexico. They use fixed effect use same Hausman/Wald/Wooldridge diagnostic which fix with the gravity model.

Brown & Hibbert (2017) examines the impact violent crime has on FDI inflow across 62 countries from 1992 to 2012. Using homicides rates as a proxy for violent crime, and their results determine that violent crime acts a deterrent to FDI. Also, mentions that crime increase the "cost of doing business" which indirectly affects a country's trade.

In case of institutional variables, Álvarez et al. (2018) examines how national institutional quality affect trade flows of 186 countries for the period pseudo-maximum likelihood estimation. Their results indicate that institutional conditions at destination country and the institutional distance between trading partners are relevant factors for bilateral trade, motivating the inclusion of governance indicators as controls.

J. E. Anderson & Marcouiller (2002) demonstrate that inadequate institutions constrain trade as much as tariffs do. They further find that omitting indices of institutional quality biases the estimates of typical gravity models, obscuring a negative relationship between per capita income and the share of total expenditure devoted to traded goods. The authors conclude that transactions costs associated with insecure exchange significantly prevent international trade.

Yu et al. (2015) analyzes bilateral trade patterns in a sample of 16 European countries between 1996-2006, finding that rules of law have positive effect on trade. When the rule of law improves the risk of non-payment, exporters face lower risks with facilitating trade.

Komissaryuk & Güngör (2026) examine the impact of the rule of law, financial inclusion in law, and economic growth on trade through econometric methods and correlation tests. Their results show a strong positive relationship between the rule of law and trade across all country groups, reinforcing the case for including institutional quality as a control variable.

In the same way, Silva & Tenreyro (2006) estimate the determinants of trade through a gravity model. They find that the GDP elasticity of trade is consistently positive and significant across OLS and PPML specifications, even when heteroskedasticity is present, as confirmed by the Wald test. The result validates the standard practice in the gravity model literature, including GDP as a primary control.

The study by Bleaney & Tian (2023) measures the ratio of GDP to trade and states that it is well-known that the trade-GDP ratio is affected by relatively time-invariant factors. Using cross-country panels by applying country fixed effects, they find that these effects of transport costs or trade policy on trade volumes

The work of Durkin Jr & Krygier (2000) examines the relationship between per capita GDP differences and bilateral trade. They find that this relationship is negative in OLS regressions but positive in fixed-effect regression, showing a positive and significant relationship between GDP per capita differences and the trade shares, especially in vertically differentiated trade. This underscores the importance of correctly specifying GDP-related controls in gravity models.

3. DATA AND METHODOLOGY

3.1. Data source

This study utilized a panel dataset consisting of 138 countries over the period 2000 to 2023. The data was primarily obtained from the World Bank's indicators (WDI). The panel is slightly unbalanced due to missing values for homicide rates in certain years, but it provides a comprehensive global view across different development levels.

Furthermore, I define the variables of the study in the next table:

Table 1: Model variable

Variable	Symbol	Definitions	Source
Log of trade	lTrade	Natural log of total export(Constant 2010). Depended variable	WDI
Homicide Rate	Homicides	International Homicides per 100,000 people. Main independent variable	WDI/UNODC
Log of GDP	lGDP	Natural log of GDP per capita (Constant 2010) controls for economic size	WDI
FDI	FDI	Foreign Direct Investment. Net inflows as a % of GDP	WDI
Rule of Low	Rules	Index of confidence in society's rules and legal system	WDI

Source: Own Elaboration

Furthermore, we provide the intuition for why each variable is relevant to estimating the effect of homicide rates on international trade flows. This approach follows the standard academic practice of grounding variable selection in the prior literature while explaining the specific economic mechanisms that motivate each choice.

In this sense, exports are the most direct measure of a country's outward economic integration and are what criminal violence plausibly harms. When violence rises, foreign buyers become more reluctant to establish trade relationships, logistic costs increase, and export firms face higher security and insurance expenses. The log transformations mean that the homicide coefficient is interpreted as a percentage change in trade per unit increase in homicides in a directly interpretable economic magnitude.

The homicide rate variable affects international trade through at least three channels. First, violence raises transaction costs directly and causes supply chain disruptions. Second, violence signals institutional weakness and unpredictability to foreign trading partners, who avoid establishing trade relationships with countries perceived as unsafe. Third, violence reduces domestic productive capacity by diverting resources toward security and deterring the domestic investment needed to produce exportable goods. The homicide rate captures all three channels simultaneously and is the best available country-level proxy for the security environment faced by international trading firms.

Regarding control, variable GDP per capita must be included as a control because wealthier, more economically developed countries tend to trade more and tend to have lower homicide rates. Without controlling for GDP, the estimated homicide coefficient would capture not just the security effect on trade, but also the fact that developing countries simultaneously have higher crime and lower trade. By including GDP, the model holds economic development constant and isolates the portion of trade variation attributed to security conditions specifically.

Foreign direct investment (FDI) is included as a control for two distinct reasons. First, FDI directly generates trade through global supply chains. When foreign firms invest in a country, they import intermediate goods and export finished products, increasing total trade volumes. A model without FDI would mistake trade generated by investment for another variable. Second, FDI is itself affected by security conditions. This creates an indirect pathway: Homicides reduce FDI, which in turn reduces trade. Including FDI as a control allows the model to separate the direct effect of homicides on trade from the indirect effect that operates through the investment channel.

The rule of law is included as a control because weak legal institutions impose costs on international trade that are separate from criminal violence. In countries with poor contract enforcement, international buyers face a greater risk of counterparty default. Critically, weak institutions generate trade costs, but this is not the same as the homicide rate: a country can have relatively low violence but weak courts or relatively high violence but functional commercial law. Without control for Rule of Law, the homicide coefficient would capture both the security effect and the institutional quality effect on trade, overstating the role of violence. The Rule of Law control isolates the portion of trade variation attributable to criminal violence net of the quality of the legal systems in which that violence occurs.

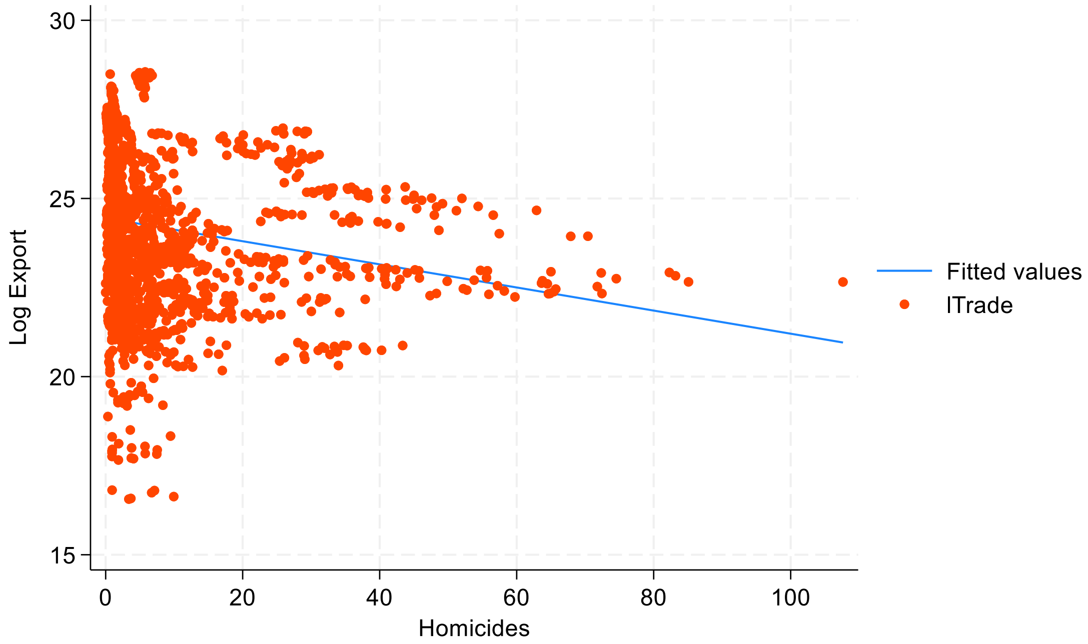
Finally, show the statistics table and graph to extend the analysis and understand the behavior of these:

Table 2. Summary statistics table

Variable	Obs	Mean	Std. dev.	Min	Max
lTrade	2,111	24.22382	2.030127	16.56254	28.55132
FDI	2,111	4.59595	29.693	-382.6866	453.1627
lGDP	2,111	9.186228	1.27151	5.829344	11.65377
Rules	2,111	.2987382	.9729499	-2.331935	2.124762
Homicides	2,111	6.921974	11.77697	0	107.6382

Source: Own Elaboration

Figure 1. Trade Flow and Homicide rate correlation



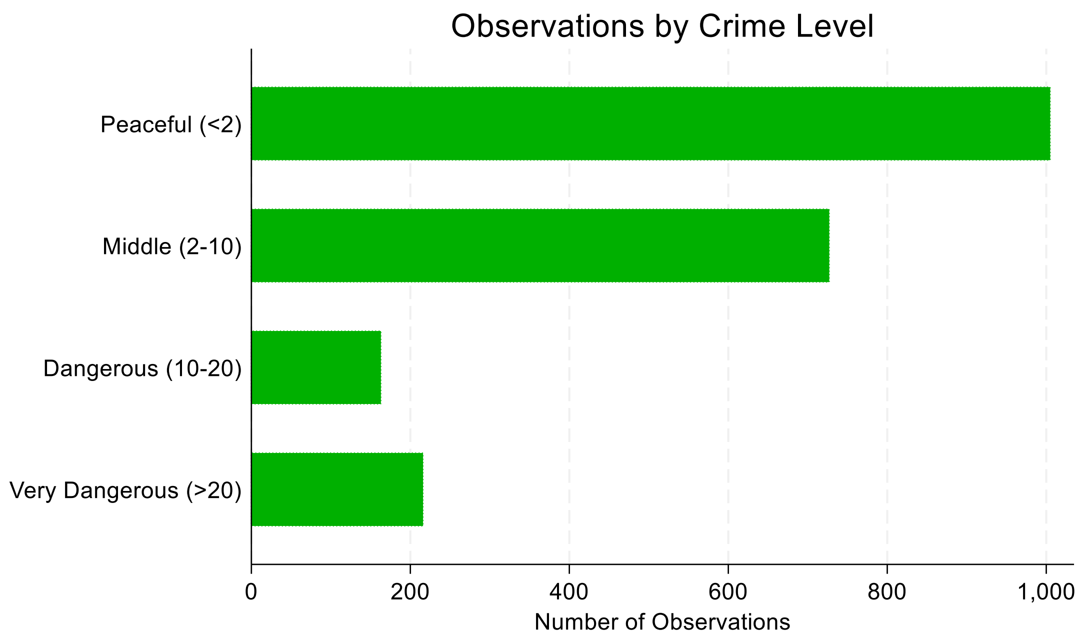
Source: Author's calculations from Work Bank WDI (2026)

Figure 1 displays the bivariate relationship between the homicide rate and log exports across all country-year observations in your panel. The blue fitted line has a clear negative slope between homicide rates and trade flows. Higher homicide rates are associated with lower trade flows at the raw, unconditional level.

The strong left-side concentrations of observations reflect the global distribution of homicides rates, with most countries clustered below 20 homicides per 100,000 inhabitants. The vast, majority of observations are clustered in this zone, reflecting the global reality: the most countries particularly high-income OECD nations, East Asian economies, and much of Europe maintain homicides rates well below 10 per 100,000 inhabitants.

On the other hand, these extreme outliers corresponding to peak-violence years in Central American countries (particularly El Salvador, Honduras and Venezuela, when rates reached 80-108 per 100,00). Importantly, theses extremes cases still show log trade values of 22-23, broadly consistent with the fitted line, suggesting they are not unduly distorting the regression.

Figure 2. Distribution of observations by Crime Level



Source: Author’s calculations from Work Bank WDI (2026)

Figure 2 shows the distributions of country-year observations across the four crime severity categories. The sample is predominantly composed of low-to-high crime environmental: approximately 47% of observations belong to Peaceful countries % (fewer than 2 homicide per 100,000) and 35 to middle crime countries (2-10 per 100,000), together accounting for 82% of the total sample.

This retribution reflects the actual global structure of homicide rates and has important implications for the analysis. First the stark imbalance across categories reinforces why a simple linear model would be inefficient because it would be overwhelmingly driven by 82% of the observations in the two lowest categories, completely masking any heterogeneity at higher violence levels. The interaction design between homicide rates and crime-level classification addresses this directly by allowing each crime level to have its own marginal effect, which is the correct approach given this skewed distribution.

3.2 Methodology

To address the research questions, a set of hypotheses is tested:

H0: Homicide rates do not affect international trade significantly across crime-level environments

H1: Homicide rates reduce international trade significantly, as violence increases

The data is structured as panel data, where countries represent cross-sectional units and years represent the time dimension. Panel data methods allow controlling for unobserved country-specific characteristics that remain constant over time.

The dependent variable is international trade, measured through exports and transformed into natural logarithms:

$$lTrade_{it} = \ln\left(Trade_{it}\right)$$

The first model estimates the contemporaneous effect of homicide rates on trade flows in static model equations

$$\begin{aligned} lTrade_{it} = \beta_0 + \beta_1 Homicides_{it} + \beta_2 lGDPp_{it} + \beta_3 FDI_{it} + \beta_4 Rules_{it} \\ + \beta_5 Crimelevel_{it} + \beta_6 Year_t + u_i + \varepsilon_{it} \end{aligned}$$

Where each variable defines:

$lTrade_{it}$ represents the natural logarithm of exports for country I in year t. It is the dependent variable and serves as a measure of international trade flows.

$Homicides_{it}$ denotes the homicide rates four countries in year t, measured as the number of international homicides per 100,000 inhabitants. It is used as a proxy for violent crime.

$lGDPp_{it}$ is the natural logarithm of Gross Domestic Product (GDP) for country I in year t. This variable controls the size and economic capacity of the country.

FDI_{it} represents Foreign Direct Investment measured as net inflows (% of GDP). It captures the degree of international capital integration and investment attractiveness.

$Rules_{it}$ corresponds to the Rule of Law indicator obtained from the Worldwide Governance Indicators. Higher values indicate stronger institutions, better contract enforcement, and greater legal certainty

$Crimelevel_{it}$ is a categorical variable that classifies countries into four groups according to their homicide rate:

Level 1: Peaceful

Level 2: Medium Crime

Level 3: Dangerous

Level 4: Very Dangerous

$Year_t$ represents year fixed effects included to control for global shocks and common time trends affecting all countries.

u_i denotes unobserved country-specific fixed effects that capture time-invariant characteristics such as geography, culture, or historical factors.

ε_{it} is the idiosyncratic error term.

Then we use dynamic specifications that include a lagged dependent variable to capture persistence in trade flow over time:

$$lTrade_{it} = \alpha L.lTrade_{it-1} + \beta_1 Homicides_{it} + \beta_2 lGDP_{it} + \beta_3 FDI_{it} + \beta_4 Rules_{it} + u_i + \varepsilon_{it}$$

$L.lTrade_{it-1}$ is the one-period lag of international trade. It captures trade persistence over time, reflecting the idea that current trade flows are influenced by past trade performance.

α measures the degree of persistence in international trade.

To determine the appropriate panel specification, the Hausman test is employed to compare fixed effects and random effects estimators.

The null hypothesis assumes that random effects are consistent:

$$H_0: Cov(X_{it}, \mu_i) = 0$$

The null hypothesis assumes that random effects are consistent:

$$H_1: Cov(X_{it}, \mu_i) \neq 0$$

The results of the Human test support the use of fixed effects models

In addition, Woodridge test and Wald test revealed the presence of serial correlations and heteroskedasticity. Therefore, robust standard errors clustered at the country level were employed to ensure reliable statistical inference.

Finally, the research includes interaction terms between homicide rates and crime-level classifications; marginal effects are estimated to facilitate interpretation.

The marginal effect of homicide rates on trade is represented as:

$$\frac{\delta lTrade}{\delta Homicides} = \beta_1 + \beta_5(CrimeLevel)$$

This expression indicates that the impact of homicide rates on trade depends on the crime environment.

To interpret these heterogeneous effects, predictive margins and marginal effects are estimated using post-estimation techniques.

4. RESULTS

Table 2. Static and Dynamic model of homicide rates on international trade flow

VARIABLES	Static FE Model	Dynamic FE Model
FDI	-0.0001	-0.0001
	(0.000)	(0.000)
lGDP	1.221***	0.392***
	(0.073)	(0.077)
Rules	0.000	-0.009
	(0.064)	(0.022)
Homicides	-0.095***	-0.023*
	(0.030)	(0.013)
yearly_crime_level = 2, Middle (2-10)	-0.232***	-0.053**
	(0.073)	(0.026)
yearly_crime_level = 3, Dangerous (10-20)	-0.309***	-0.059
	(0.109)	(0.077)
yearly_crime_level = 4, Very Dangerous (>20)	0.091	0.025
	(0.126)	(0.044)
2.yearly_crime_level#c.Homicides	0.108***	0.026*
	(0.034)	(0.013)
3.yearly_crime_level#c.Homicides	0.114***	0.028**
	(0.031)	(0.013)
4.yearly_crime_level#c.Homicides	0.097***	0.025*
	(0.030)	(0.013)
lTrade = L,		0.729***
		(0.050)
Constant	12.896***	3.003***
	(0.665)	(0.730)
Observations	2,111	1,815
R-squared	0.811	0.917
Number of country_id	138	124

Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0

Source: Author’s calculations from Work Bank WDI (2026)

We present the regression table where the present Static Fixed model, which R^2 explains 81.1% of the relation within-country variation in log trade, with control variables accounting for most of the trade variation across countries and years.

In the same way, the Dynamic Fixed model with R^2=0.92 is driven mainly by the lagged trade term, which gives us a good fit of the model. The 296-observation reduction is expected because one year of the data per country is consumed by the lag operators.

Both models are estimated with cluster-robust standard errors, addressing the heteroskedasticity and serial correlation confirmed by the modified Wald test and the Wooldridge test. This has similarities with Garriga & Phillips (2023); Pan & Xu (2025) Garriga & Phillips, (2023); Pan & Xu, (2025) who used tests to detect autocorrelation.

The results show that the homicide coefficient captures the base line effect of a one-unit increase in the homicide rate on trade flows for countries, also taking into account the reference category, peaceful countries.

Therefore, in our Static model, a one-unit increase in the homicide rate is associated with a 9,5% decrease in trade flows, which is statistically significant for a peaceful country, holding all other variables constant. This outcome is consistent with Bandyopadhyay & Sandler (2014), who find that violence is equivalent to a 30% tariff on bilateral trade, and with M. A. Anderson et al. (2022), who find that homicide rates significantly reduce export capacity, especially in differentiated manufactured and services.

Likewise, Dynamic model indicates that in the short run, the coefficient on homicides affects 2.3% this statistically significant to 10%. According to the model Trade is highly persistent in the variable Ltrade=L (rho=0,729) over time, in the most precisely estimated coefficient in the dynamic model, confirming that international trade in characterized by strong inertia.72,9% of country's previous year's trade volume persists into the current year, independent of current economic conditions, so not all homicide effects manifest immediately.

This persistence reflects the structural features of international trade: sunk market entry cost. The high persistence coefficient validates the dynamic specification: If LlTrade were not significant, the dynamic extension would be unnecessary. Its strong significance ($p<0.01$) confirms that omitting it from the static model introduces dynamic misspecification, and the model provides a more complete picture of trade adjustment. This is consistent with the study of Loncan (2024), which, through dynamic regression, confirms that violence reduces trade.

Then using Arellano and Bond (1993), which is a long-run formula in the panel -0.023/(1-0729) to get the long run in the model. The effect is -8.5%, which converges closely with the static estimate of 9.5% proving strong confirmation that both models are capturing the same underlying structural relationship. This convergence is a key robustness result because the static model can be interpreted as approximating the long-run equilibrium.

In the other hand, we measure Crime Level Category Effects. The dummies measure the structural trade difference between countries at each crime category, where the base group continue being Peaceful countries. In this sense, countries classified as middle crime trade 23,2% less than Peaceful countries, statistically significant, and 5.3% less in the dynamic model. This is a robust structural penalty, reflecting cumulative institutional, reputational, and logistical costs associated with operating un a moderately insecure environment. This is coherent outcome of Rozo, (2018)

In dangerous countries, the static trade penalty deepens to 30.9%, statistically significant, the largest baseline reduction across all categories. This suggests the 10-20 homicide rate represents a critical threshold where systematic insecurity disrupts trade. The dynamic coefficient shows 5,9% but it is not significant, that indicating that once trade persistence is controlled, within country variation in this classification lacks sufficient precision for the short-run dynamic estimate. Rutar (2026) in their study has resemblance to the analysis he determines there is not robust evidence in global panel and states that the relation between violence and crime are fragile.

To continue, the coefficient for Very Dangerous countries is positive 9,1% static and 2.5% in the dynamic model, but statistically insignificant. This apparent result is a well-documented phenomenon in nonlinear crime-trade literature and reflects several mechanisms. For instance, countries with chronic extreme violence tend to be commodity-dependent economies (oil, minerals) whose trade is driven by global commodity demand, not domestic security conditions. Similarity to Ma et al. (2025); Rutar (2026) research.

Also, firms operating in persistently extreme violence environments have developed informal security arrangements and alternative supply chains, partially insulating trade from marginal increases in homicides. Although in the study, the very dangerous category likely contains fewer country-year observations, reducing estimation precision.

As a control variable, this research uses the logarithm of GDP as a main economic driver. In this sense, GDP is a dominant control variable, producing robust results in both specifications. In the static model, a 1% increase in GDP is associated with 122% increase in trade. Which means a super -unitary elasticity consistent with the standard gravity model.

Besides, in the dynamic specification, the coefficient increases to approximately 39%. This reflects that part of the GDP effect is transmitted through the lagged trade variable. Consequently, past trade itself is partly a product of high GDP, so the direct GDP coefficient in the dynamic model captures only the additional short-run effect after controlling for trade persistence. This result is consistent with the study of Silva & Tenreyro (2006) and Bleaney & Tian (2023), where, by applying country fixed effects, they got significant effects on trade.

Another control variable, FDI, is statistically insignificant in both models. This result, while unexpected given the theoretical complementarity between FDI and trade, is common in Fixed Effects panel estimations. The FE estimators identify only within-country variation over time; much of the FDI-trade relationship reflects stable, country-level structural factors that are absorbed by country fixed effects. FDI may also affect trade through long-run channels not captured by contemporaneous coefficients. These results are not coincident with Loncan (2024) who discover negative association between homicides, investment and trade.

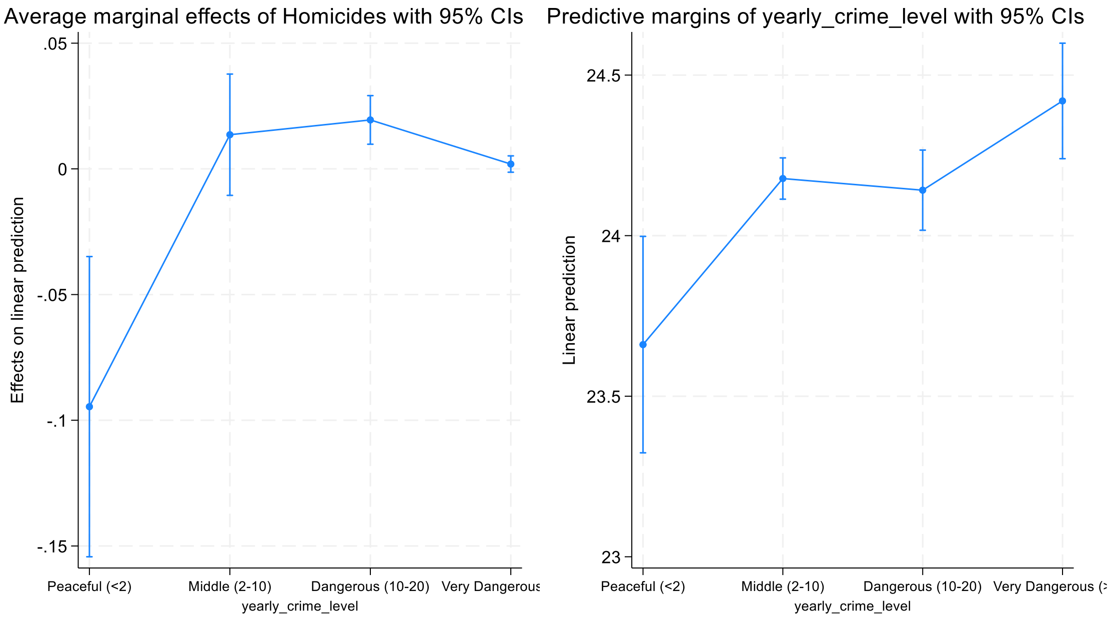
The last control variable, the Rule of Law index, is statistically insignificant in both specifications. Institutional quality measures change very slowly within countries over time, producing low within-country variance. The bulk of cross-national institutional differences is absorbed by country fixed effects. This does not imply institutions are important for trade. Cross-sectional evidence firmly establishes their importance. It simply means that in this panel, time-varying changes in the Rule of Law are insufficient to generate a precisely estimated within-country effect. Contrary to Álvarez et al. (2018; Yu et al. (2015), and Komissaryuk & Güngör (2026), who found a strong positive relationship between the rule of law and trade in all groups of countries

To conclude, we need to study the interaction terms (Homicides##.year) test, where the marginal effect points out that an additional homicide or unit of crime on trade differs across crime severity categories

Marginal effect allows evaluating how the impact of homicide rates on international trade varies across different levels of crime intensity. This approach follows standard econometric practices for interpreting interaction terms in nonlinear models (Williams, 2012; Wooldridge, 2010).

4.1. Margin effect in the static and dynamic model

Figure 3. Margin effect of static and dynamic panel data



Source: Author’s calculations from Work Bank WDI (2026)

Figure 3 presents the post-estimation marginal effects of the static and dynamic models. The panel left reports the marginal effects of homicide trade rates across crime categories. The coefficient in the peaceful countries group is -0.0946; the average marginal effects mean the strongest and most precisely estimated effect. In other words, a one-unit increase in homicide rates reduces trade by 9.46%. The coefficient interval is entirely negative, confirming that the trade-damaging effect is robust for this group. For middle-crime countries, the coefficient is -0.0136. The effect is statistically indistinguishable from zero. In this case, we are confirming no reliable marginal effect at this crime level. Therefore, for countries in the 2-10 homicides range, additional violence neither significantly reduces nor increases trade. This is the first critical threshold in the crime trade relationship.

The Dangerous Countries Group presents an average marginal effect of 0.0195. This result does not imply that violence promotes trade. It most likely reflects a compositional effect: countries driving within-country variation in homicide rates at this level may simultaneously be experiencing trade expansion from commodity booms, urbanization, or regional integration. Alternatively, it may reflect measurement limitations of using aggregate trade rather than differentiated-goods trade as the dependent variable.

In the case of very dangerous countries, the coefficient +0.0019 is near-zero and statistically insignificant. The coefficient intervals straddle zero, indicating no reliable relationship between marginal homicide increases and trade at the highest violence level. Therefore, at extreme violence levels, the most trade-sensitive activities have already been eliminated, and remaining trade is insensitive to further violence.

The second panel is a dynamic model, with 95% confidence intervals. From the Dynamic model at each crime level, evaluated at the sample means of all continuous covariates. They show the expected trade level, not the change in trade.

The predicted log trade values are remarkably similar across crime levels: 23,66 to Peaceful countries, 14.18 for Middle, 24.14 for Dangerous, and 24.42 for Very Dangerous. All four predicted values are highly statistically significant ($p<0.001$ in all cases), confirming that all four crime-level groups have well-identified, precisely estimated predicted trade levels in the dynamic model.

Importantly, the Peaceful category shows the lowest predicted trade levels (12.66), while the Dangerous countries show (24.42). This does not mean violence increases trade. These predictions are commodity-rich emerging economies (Brazil, South Africa, Honduras, Venezuela) with substantial aggregate trade volumes driven by factors unrelated to security. The margin captures the trade level of the average country within each category, not the causal effect of moving between categories. These results are consistent with the study of M. A. Anderson et al. (2022); and Morales & Ricca, (2019).

5. CONCLUSION

The empirical findings provide evidence that violent crime negatively affects international trade. In the static model, an increase in homicide rates is associated with a statistically significant reduction in trade flows. The dynamic specification confirms this result after controlling trade persistence through the inclusion of the lagged dependent variable. Although the magnitude of the homicide coefficient decreases in the dynamic model, the negative relationship remains robust, suggesting that crime imposes long-term costs on international economic activity.

The results also reveal important heterogeneity across crime environments. Countries classified as middle-crime and dangerous exhibit significant trade penalties relative to peaceful countries in the static specification. These findings suggest that insecurity generates additional transaction costs, increases uncertainty, and discourages participation in international markets. By contrast, the very dangerous category does not display a statistically significant additional penalty. These results may reflect the presence of commodity-exporting economies whose trade performance depends more strongly on external demand than on domestic security conditions.

The marginal effects analysis provides further insights into the crime-trade relationship. The strongest negative marginal effects of homicide rates are observed among peaceful countries, where additional increases in violence are associated with substantial reductions in trade. However, the marginal effects become statistically weaker at high levels of crime. This pattern suggests that trade is particularly sensitive to increases in violence when countries begin with relatively secure environments. Once high levels of violence become persistent, economic agents may adapt through alternative security arrangements, informal networks, or sectoral specialization, reducing the sensitivity of trade to additional increases in crime.

Among the control variables, GDP per capita emerges as the most important determinant of trade flows. The positive and statistically significant GDP coefficient is consistent with the gravity model literature, showing that larger economies tend to trade more. Foreign Direct Investment and Rule of Law do not achieve statistical significance in the fixed-effects estimations. This result likely reflects the limited within-country variation of these variables over time rather than a lack of economic relevance.

In general, the findings support the view that crime constitutes an economic barrier to international trade. Beyond the direct social cost associated with violence, insecurity can reduce competitiveness, increase transaction costs, and weaken integration into global markets. Consequently, security improvements generate the highest trade returns in countries that are already relatively peaceful. In high-violence countries. Marginal reductions in homicide rates appear to have limited direct trade effects so economic reforms are required alongside security improvements to restore capacity.

In conclusion, the evidence presented in this thesis indicates that violent crime is an important determinant of international trade performance. While the magnitude of the effect varies across crime environments, the overall results consistently suggest that higher levels of violence reduce trade and hinder economic integration. These finding contribute to the growing literature linking security conditions to economic development and international commerce.

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7. APPENDIX

7.1. Additional statistical results

This appendix presents the supplementary statistical output used to support the empirical analysis. These materials are included for transparency and replication purposes but are not discussed in detail in the main text.

Figure 4. Descriptive Statistics

Variable		Mean	Std. dev.	Min	Max	Observations
lTrade	overall	24.22382	2.030127	16.56254	28.55132	N = 2111
	between		2.238305	16.68919	28.49057	n = 138
	within		.3155638	22.55751	25.35216	T-bar = 15.2971
FDI	overall	4.59595	29.693	-382.6866	453.1627	N = 2111
	between		12.46994	-5.130038	97.04396	n = 138
	within		25.84305	-474.6757	360.7147	T-bar = 15.2971
lGDP	overall	9.186228	1.27151	5.829344	11.65377	N = 2111
	between		1.364261	6.084726	11.57087	n = 138
	within		.1654312	8.24859	9.702642	T-bar = 15.2971
Rules	overall	.2987382	.9729499	-2.331935	2.124762	N = 2111
	between		.9406327	-1.738332	1.976096	n = 138
	within		.1640947	-.572897	1.171836	T-bar = 15.2971
Homici~s	overall	6.921974	11.77697	0	107.6382	N = 2111
	between		9.559356	.2471851	55.51867	n = 138
	within		3.965784	-40.69901	59.04153	T-bar = 15.2971

Source: Author’s calculations from Work Bank WDI (2026)

Figure 5. Correlation matrix

```
. pwcorr lTrade  Homicides FDI lGDP Rules , star(.01) bonferroni

             |   lTrade Homici~s      FDI     lGDP    Rules
-------------+---------------------------------------------
      lTrade |   1.0000 
   Homicides |  -0.1882*  1.0000 
         FDI |   0.0341  -0.0663   1.0000 
        lGDP |   0.6023* -0.2676*  0.1375*  1.0000 
       Rules |   0.4976* -0.4111*  0.1425*  0.8349*  1.0000 
```

Source: Author's calculations from Work Bank WDI (2026)

Figure 6. Hausman test: Choosing between a fixed and random effects

```
. xtreg lTrade Homicides lGDP FDI Rules i.yearly_crime_level lGDP FDI Rules , fe
note: lGDP omitted because of collinearity.
note: FDI omitted because of collinearity.
note: Rules omitted because of collinearity.

Fixed-effects (within) regression               Number of obs     =      2,111
Group variable: country_id                      Number of groups  =        138

R-squared:                                      Obs per group:
     Within  = 0.7541                                         min =          1
     Between = 0.3898                                         avg =       15.3
     Overall = 0.3639                                         max =         23

                                                F(7,1966)         =     861.25
corr(u_i, Xb) = -0.4475                         Prob > F          =     0.0000

------------------------------------------------------------------------------------------
                  lTrade | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------------------+----------------------------------------------------------------
               Homicides |  -.0000304   .0010554    -0.03   0.977    -.0021002    .0020394
                    lGDP |   1.664504   .0228378    72.88   0.000     1.619715    1.709293
                     FDI |   .0000278   .0001366     0.20   0.839    -.0002401    .0002958
                   Rules |  -.1245745   .0227923    -5.47   0.000    -.1692741   -.0798749
                         |
      yearly_crime_level |
           Middle (2-10) |  -.0908143   .0174818    -5.19   0.000    -.1250991   -.0565294
       Dangerous (10-20) |  -.0336251   .0281893    -1.19   0.233    -.0889092     .021659
    Very Dangerous (>20) |   .1103706   .0442421     2.49   0.013     .0236042     .197137
                         |
                    lGDP |          0  (omitted)
                     FDI |          0  (omitted)
                   Rules |          0  (omitted)
                   _cons |   8.993188   .2113075    42.56   0.000     8.578778    9.407598
-------------------------+----------------------------------------------------------------
                 sigma_u |  1.9216999
                 sigma_e |  .16211634
                     rho |  .99293353   (fraction of variance due to u_i)
------------------------------------------------------------------------------------------
F test that all u_i=0: F(137, 1966) = 1479.24               Prob > F = 0.0000
```

Source: Author's calculations from Work Bank WDI (2026)

Figure 7. Hausman test: Choosing between fixed and random effects

```
. xtreg lTrade Homicides lGDP FDI Rules i.yearly_crime_level lGDP FDI Rules, re
note: lGDP omitted because of collinearity.
note: FDI omitted because of collinearity.
note: Rules omitted because of collinearity.

Random-effects GLS regression                   Number of obs     =      2,111
Group variable: country_id                      Number of groups  =        138

R-squared:                                      Obs per group:
     Within  = 0.7541                                         min =          1
     Between = 0.3901                                         avg =       15.3
     Overall = 0.3640                                         max =         23

                                                Wald chi2(7)      =    6012.88
corr(u_i, X) = 0 (assumed)                      Prob > chi2       =     0.0000
```

lTrade	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
Homicides	-.0001367	.0010613	-0.13	0.897	-.0022168	.0019433
lGDP	1.643743	.0226305	72.63	0.000	1.599388	1.688098
FDI	.000027	.0001375	0.20	0.844	-.0002424	.0002964
Rules	-.1332401	.0228146	-5.84	0.000	-.1779559	-.0885243
yearly_crime_level						
Middle (2-10)	-.0941223	.0175687	-5.36	0.000	-.1285564	-.0596882
Dangerous (10-20)	-.0403359	.0283258	-1.42	0.154	-.0958534	.0151816
Very Dangerous (>20)	.1031	.0444719	2.32	0.020	.0159368	.1902633
lGDP	0	(omitted)				
FDI	0	(omitted)				
Rules	0	(omitted)				
_cons	9.227278	.2524771	36.55	0.000	8.732432	9.722124
sigma_u	1.7508766					
sigma_e	.16211634					
rho	.99149968	(fraction of variance due to u_i)				

Source: Author's calculations from Work Bank WDI (2026)

Figure 8. Hausman test: Choosing between fixed and random effects

```
. hausman fixed random

                 ---- Coefficients ----
             |      (b)          (B)            (b-B)     sqrt(diag(V_b-V_B))
             |     fixed        random       Difference       Std. err.
-------------+----------------------------------------------------------------
   Homicides |   -.0000304    -.0001367        .0001063               .
        lGDP |    1.664504     1.643743        .0207607         .0030702
         FDI |    .0000278      .000027        8.20e-07               .
       Rules |   -.1245745    -.1332401        .0086656               .
yearly_cri~l |
          2  |   -.0908143    -.0941223         .003308               .
          3  |   -.0336251    -.0403359        .0067108               .
          4  |    .1103706        .1031        .0072706               .
------------------------------------------------------------------------------
                          b = Consistent under H0 and Ha; obtained from xtreg.
           B = Inconsistent under Ha, efficient under H0; obtained from xtreg.

Test of H0: Difference in coefficients not systematic

    chi2(7) = (b-B)'[(V_b-V_B)^(-1)](b-B)
            =  39.49
Prob > chi2 = 0.0000
(V_b-V_B is not positive definite)
```

Source: Author’s calculations from Work Bank WDI (2026)

Figure 9. Wooldridge test- Autocorrelation measure

```
. do "C:\Users\pablo\AppData\Local\Temp\
> STD8420_000000.tmp"

. xtserial lTrade Homicides lGDP FDI Rul
> es                     //Preset auto//

Wooldridge test for autocorrelation in p
> anel data
H0: no first-order autocorrelation
    F(  1,     117) =    157.092
           Prob > F =      0.0000
```

Source: Author’s calculations from Work Bank WDI (2026)

Figure 10. Wald y Breusch-Pagan Test-Detention of heteroscedasticity

```
Modified Wald test for groupwise heteroskedasticity
in fixed effect regression model

H0: sigma(i)^2 = sigma^2 for all i

chi2 (138)  =      70020.93
Prob > chi2 =          0.0000


.
.
end of do-file
```

Source: Author's calculations from Work Bank WDI (2026)